



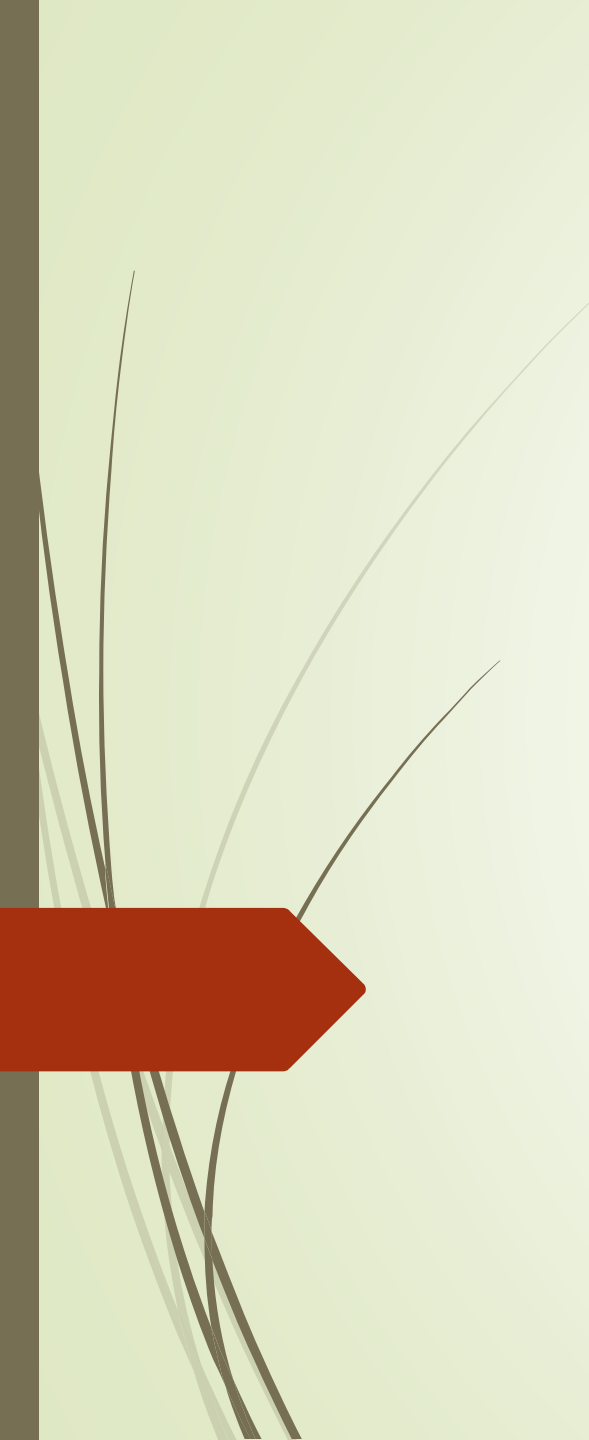
23AID202

Introduction to Robotics

Dr. Divya Udayan J, PhD(Konkuk University, S.Korea)
Associate Professor
Dept. of Computer Science & Engineering
Amrita Vishwa Vidyapeetham

Amrita Vishwa Vidyapeetham
Amritapuri Campus





Introduction to Robotics

Robot Specifications or Characteristics

1. Number of axes,
2. Load carrying capacity, kg,
3. Maximum speed, mm/s,
4. Reach and Stroke, mm,
5. Tool orientation, deg,
6. Repeatability, mm,
7. Precision and Accuracy, mm,
8. Operating environment.

Rhino XR-3 Robot Specifications (contd.)

Characteristic	Value	Units
Number of axes	5	—
Load-carrying capacity	0.5	kg
Maximum tool-tip speed	25.0	cm/sec
Horizontal reach and stroke	62.23	cm
Vertical reach and stroke	88.27	cm
Tool pitch range	270.0	deg
Tool roll range	infinite	deg
Repeatability	± 0.1	cm

Rhino XR-3 Robot Specifications (contd.) Load Shaft Precision

Motor	Joint	Precision
F	Base	0.2269
E	Shoulder	0.1135
D	Elbow	0.1135
C	Tool pitch	0.1135
B	Tool roll	0.3125

Units of precision: degrees per count.

Robot axes

Robot axes (or degrees of freedom) are the independent joints that allow a robotic arm to move. They are broadly split into positioning axes (to move the tool to a specific location in 3D space) and orientation axes (to angle the tool precisely).

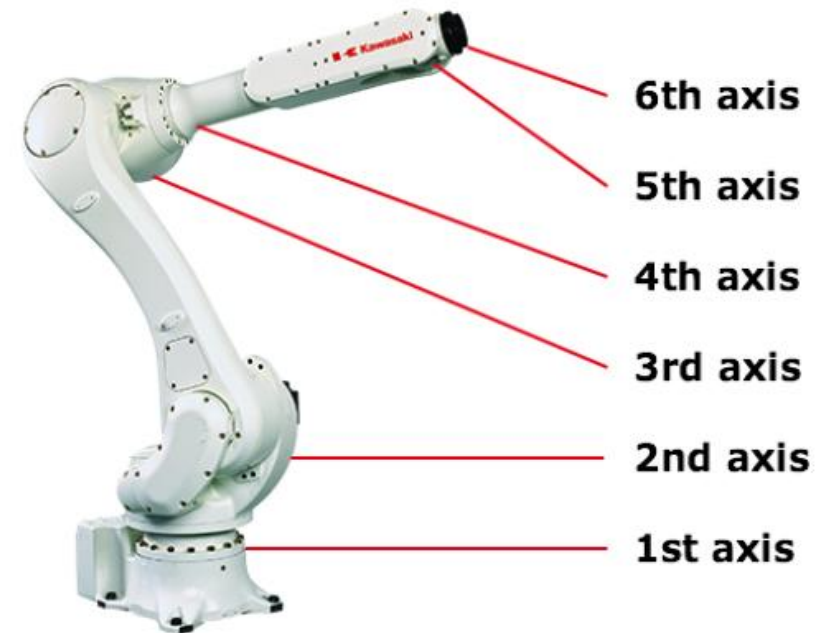
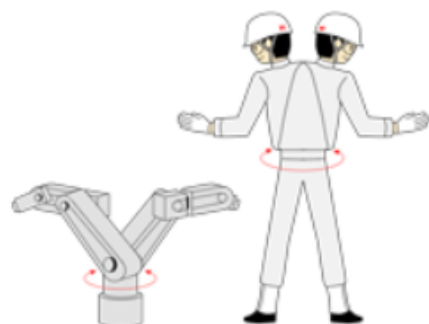
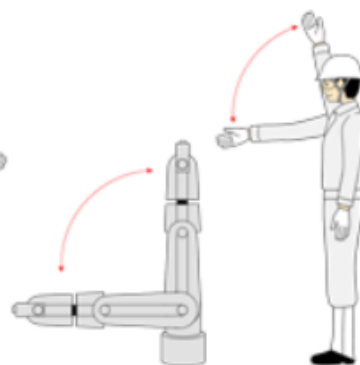
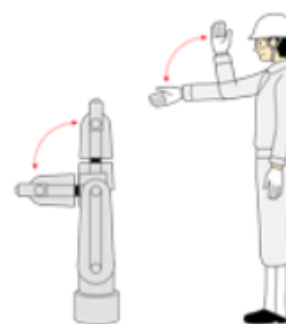
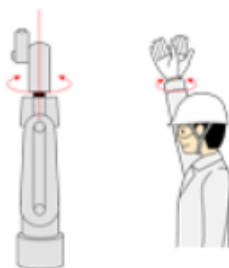
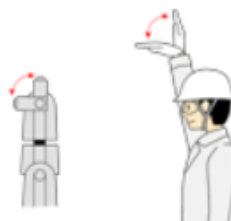


Figure 1: Six axes of motion for a vertical articulated industrial robot.

(Image: **Kawasaki**)

1st axis: Waist**2nd axis: Shoulder****3rd axis: Elbow****4th axis: Wrist (Rotating)****5th axis: Wrist (Bending)****6th axis: Fingertip**

Comparison between robot and human movement. (Image: **Kawasaki**)

Position Axes (The "Arm")

- **Axis 1 (Base/Swivel):** Rotates the entire robot left or right, determining the working range and horizontal sweep.
- **Axis 2 (Shoulder):** Controls the forward and backward pitching of the lower arm, allowing the robot to reach forward or adjust its height.
- **Axis 3 (Elbow):** Moves the upper arm up and down, extending the robot's vertical and horizontal reach. [www.atiny-lasersensor....](#) +2

Orientation Axes (The "Wrist")

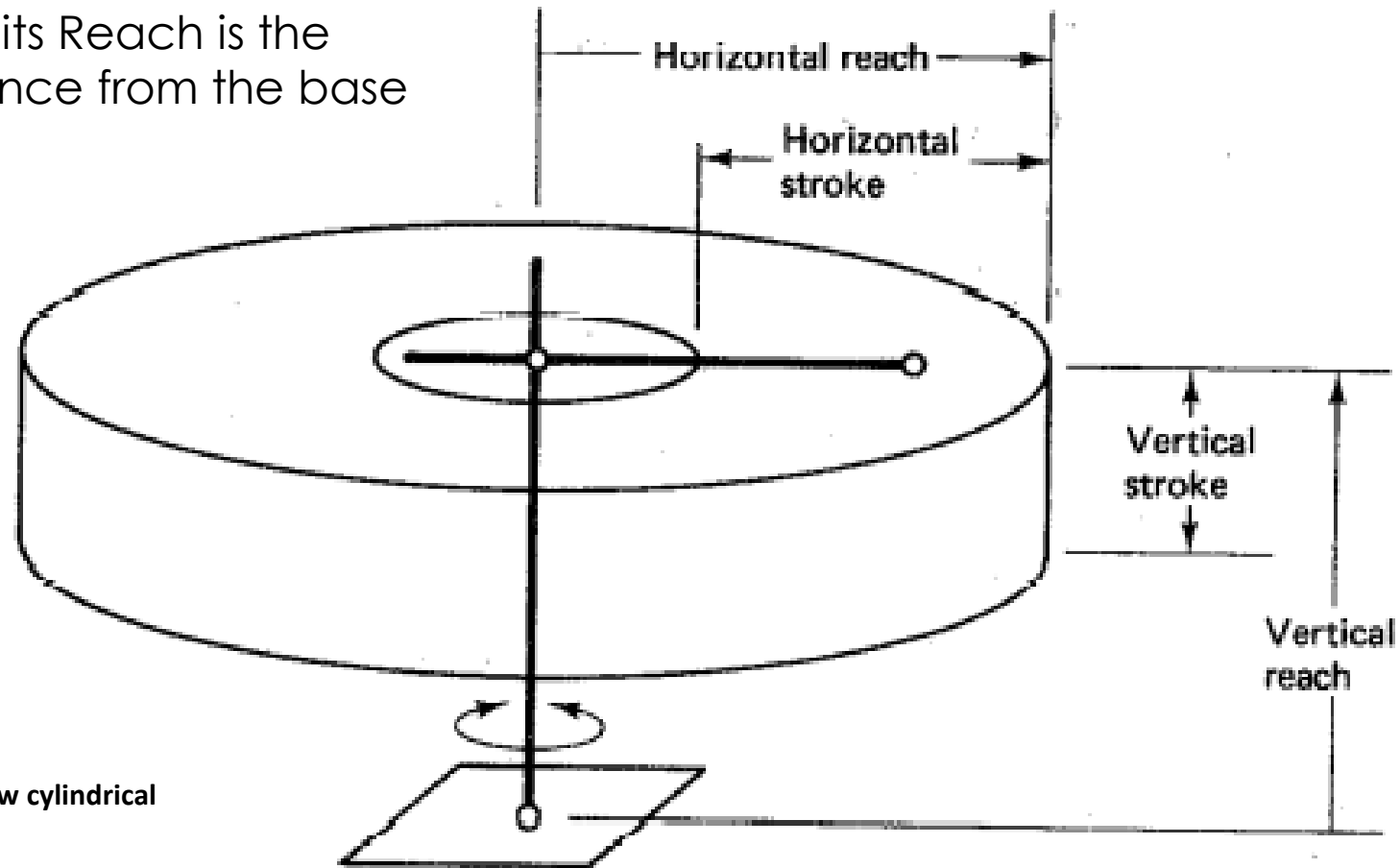
- **Axis 4 (Wrist Roll):** Twists the forearm, allowing tools like grippers or welding torches to rotate.
- **Axis 5 (Wrist Pitch/Bend):** Pivots the tool up and down to approach parts at various angles.
- **Axis 6 (Wrist Twist/Turn):** Rotates the faceplate at the end of the robot to fine-tune the final position of the attached tool. [YouTube · Will Automate](#) +4

Function of Robot Axes

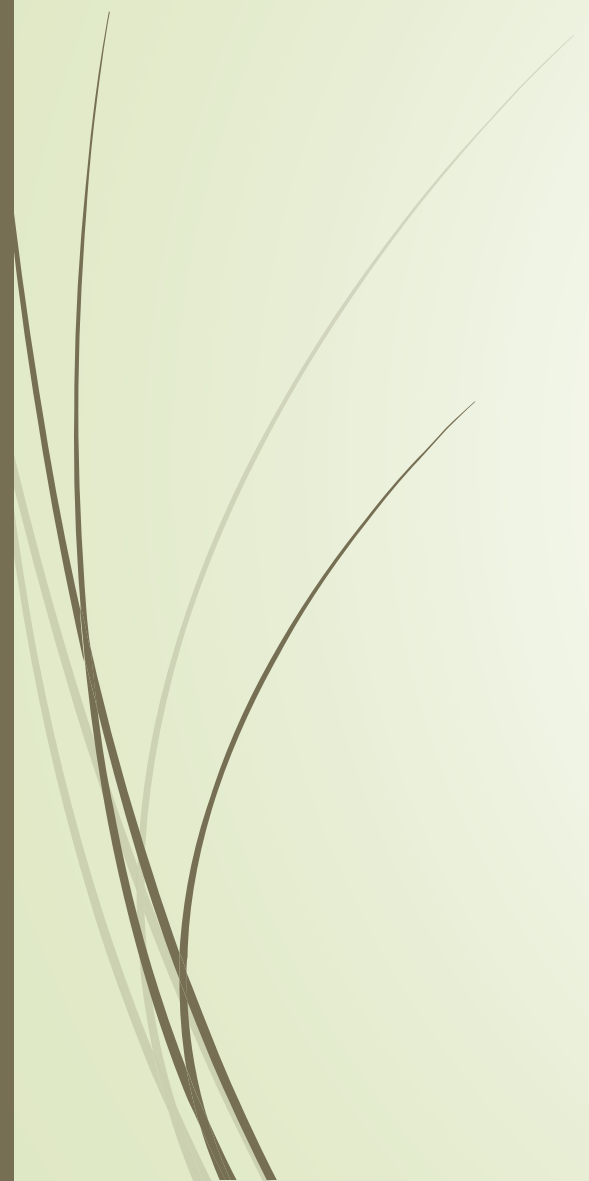
Axes	Type	Function
1-3	Major	Position the wrist
4-6	Minor	Orient the tool
7- n	Redundant	Avoid obstacles

Reach and Stroke of a Cylindrical robot

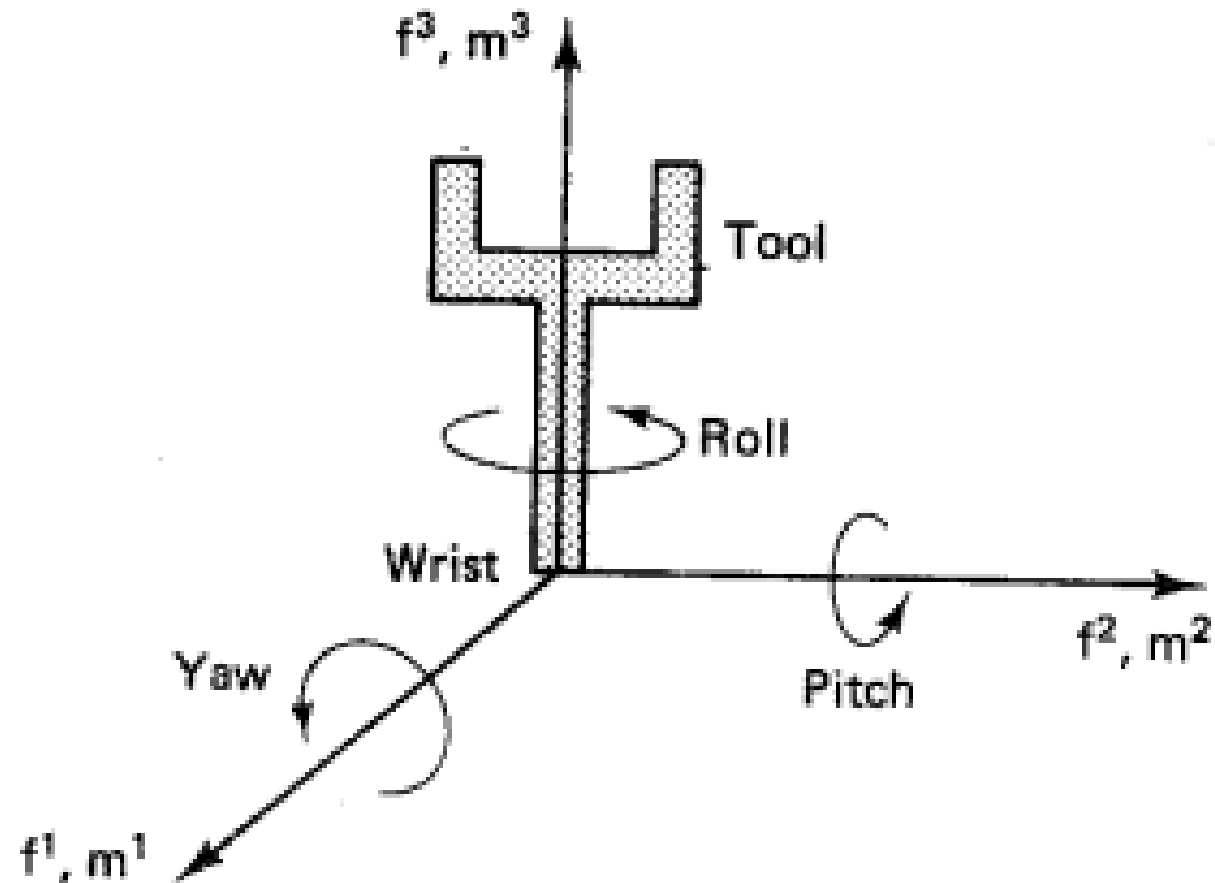
A cylindrical robot uses two linear strokes and one rotary joint. Its Stroke (radial in/out and vertical up/down) governs the dimensions of its telescoping arm, while its Reach is the maximum horizontal distance from the base



The combination of these movements creates a hollow cylindrical work envelope.



Tool Orientation (Yaw-Pitch-Roll or X-Y-Z system)



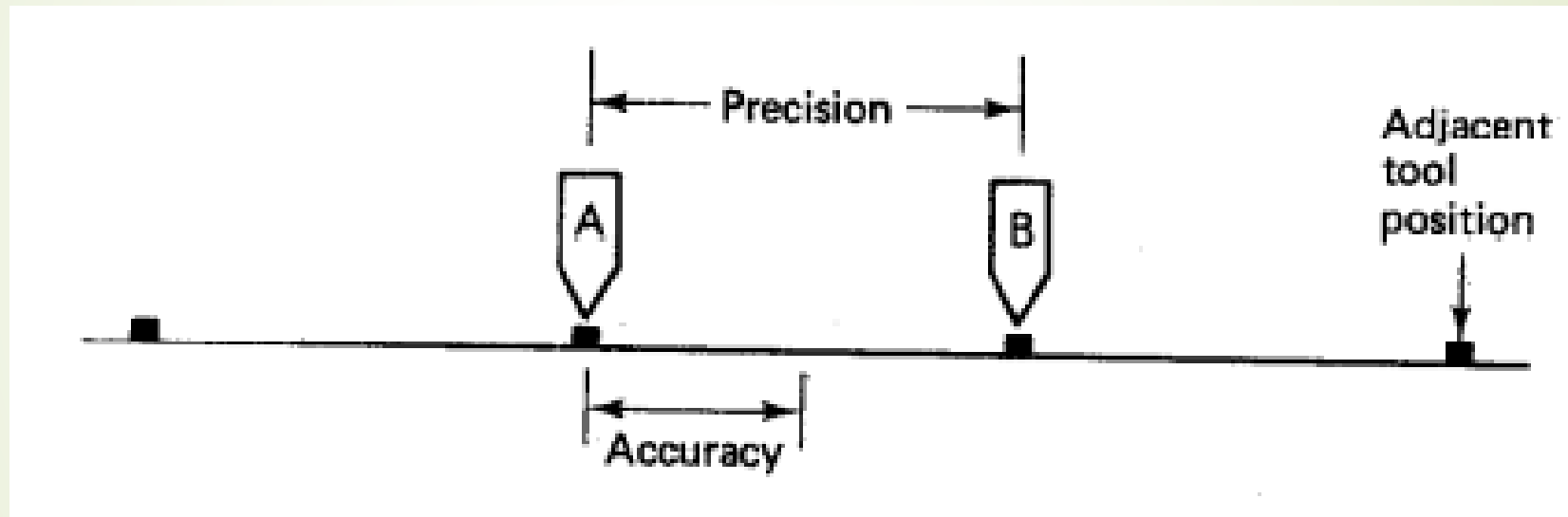
Repeatability, Precision and Accuracy

Repeatability is a measure of the ability of a robot to position the tool tip in the same position repeatedly.

Precision of a robot is the spatial resolution with which the tool can be positioned within the work envelope.

Accuracy of a robot is a measure of the ability of the robot to place the tool tip at an arbitrarily prescribed position in The work envelope.

Repeatability, Precision and Accuracy (contd.)



$$\text{Error} \geq \frac{\text{precision}}{2}$$

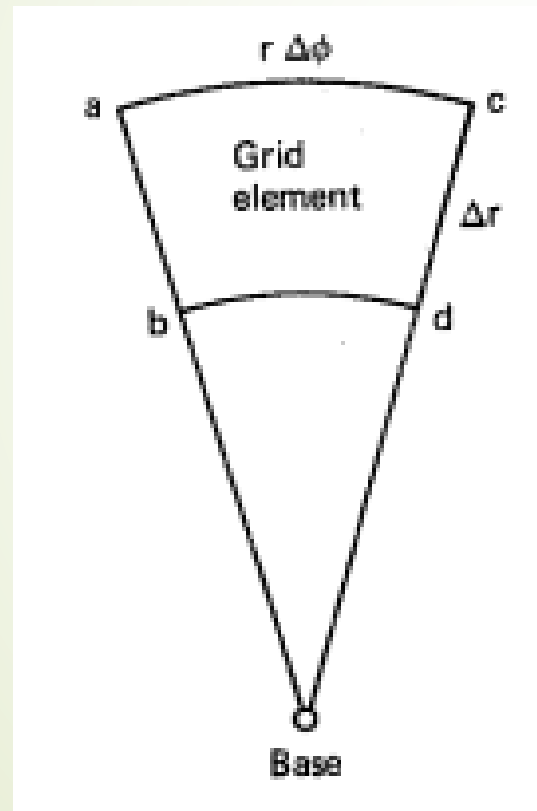
Note: Error depends on:

1. gear backlash,
2. elastic deformations in links due to loading.

Repeatability, Precision and Accuracy (contd.)

Type of Robot	Vertical Precision	Horizontal Precision	Repeatability	Accuracy
Rectangular	Uniform	Uniform	Very Good	Very Good + Highly Accurate
Cylindrical	Uniform	Decreasing radially	Good	Good + Accurate
Spherical	Decreasing radially	Decreasing radially	Satisfactory	Less Accurate
SCARA	Uniform	Decreasing radially	Very Good	Very Good
Articulate	Decreasing radially	Decreasing radially	Good	Very, Very Less Accurate

Horizontal Precision of a Cylindrical Robot



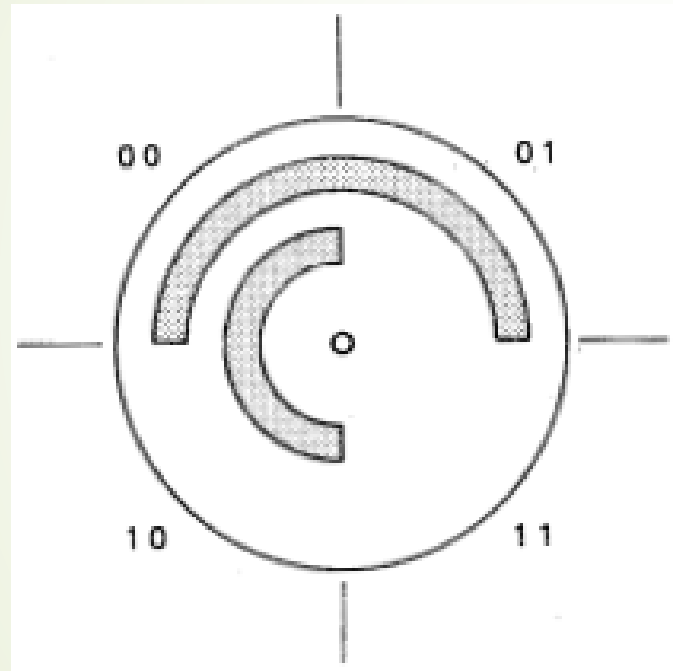
$$\Delta h \approx [(\Delta r)^2 + (R \Delta \phi)^2]^{1/2}$$

$$\Delta p \approx [(\Delta r)^2 + (R \Delta \phi)^2 + (\Delta z)^2]^{1/2}$$

$$\Delta \phi_{DAC} = \frac{\phi_{max} - \phi_{min}}{2^n} \text{ radians per count}$$

where, Δh = overall horizontal precision,
 Δp = total robot precision,
 $\Delta \Phi$ = load shaft precision.

Slotted Disk with 2-bit Gray Code



$$\Delta\phi_{inc} = \frac{\pi}{km 2^{d-1}} \text{ radians per count}$$

where, $\Delta\Phi$ = overall incremental
load shaft precision,
 m = gear reduction ratio,
 k = number of slots around
the circumference of
the disk,
 d = number of emitter-detector
pairs.

Operating Environment

- 1. Harsh, dangerous and unhealthy environments such as transport of radioactive materials, spray painting, welding, loading and unloading of furnaces, etc.**
- 2. Clean rooms are required for semiconductor industry, where temperature, humidity and airflow are controlled.**